

Usman Akhter – 24799861

04/04/2025

Digital Forensics Report

CASE NO: 023

Investigators details:

INVESTIGATOR	Usman Akhter - 24799861
OCCUPATION	Jr SOC Analyst at [REDACTED]
CASE NO	023
CASE TITLE	National Art Gallery Incident

Suspect details:

SUSPECT	TRACY
OCCUPATION	Supervisor
REASON FOR ARREST	Conspiracy to commit theft
RELEVANT DETAILS	Coral (Tracy's Alias) coralbluetwo@hotmail.com tracysuntweleve@gmail.com

Table of Contents

INTRODUCTION.....	4
1.1 Purpose and Scope of this Report.....	4
1.2 Synopsis	4
1.3 Structure and Contents	4
1.4 ACPO Principles.....	5
INVESTIGATION.....	7
2.1 Items Analysed	7
2.2 Integrity Preservation	8
2.3 Methodology	9
2.4 Investigation Results and Analysis	10
2.4.1 Stamp Insurance	10
2.4.2 Web search	13
2.4.3 Coral Alias.....	14
2.4.3 Encrypted documents	15
2.4.4 Use of Virtual Machine	16
2.4.5 Breach of Confidentiality	17
CONCLUSION.....	19
REFERENCES	20
APPENDIX A	21
APPENDIX B.....	23

INTRODUCTION

1.1 Purpose and Scope of this Report

The Purpose of this report is to conclude my Digital Forensic Investigation regarding National Art Gallery incident where police seized number of devices including external hard drive belonging to Tracy who was taken into custody on conspiracy to commit theft. The main objective of this report is to examine the disk image of Tracy's external hard drive this would reveal the evidence needed to prove Tracy is guilty in court for attempting to commit theft in National Art Gallery by stealing valuable rare stamps as well as providing assistance to Carry for carrying out a flash mob. The disk image will be analysed through using Autopsy on Windows 11

1.2 Synopsis

Tracy is a Supervisor at National Art Gallery she knows Carry who contacts Tracy and is looking to organize a flash mob at the Gallery through help of Tracy this could be by bringing suspicious items but are not illegal and giving sensitive information that could help Carry. Carry has agreed to give money to Tracy for the help, Tracy is struggling with money as she is divorced recently, and her child goes to expensive private school.

Tracy is also in contact with her brother about stealing specific stamps that are going to be on display from the National Art Gallery that are considered rare and valuable.

Joe who is Tracy's ex-husband installed key logger onto her computer before the divorce to monitor their daughter online activity this led to leaking classified information about committing theft and which then Joe alerted the authorities and then police has taken custody of various devices including Tracy's hard drive.

1.3 Structure and Contents

This report will provide support evidence and detailed findings from forensics process using Autopsy in investigations sections I will provide screenshots of emails/messages that proves Tracy was involved in conspiracy to commit theft.

This report will also contain conclusions to summarise the evidence from the viewpoint of expert witness.

1.4 ACPO Principles

ACPO guidelines provide basic guidance for handling digital evidence in law enforcement investigations these guidelines involve regarding best practices for seizing, acquisition and examining digital evidence. These guidelines contain four key principles that every digital forensics investigator should adhere to during the investigation. Horsman, G. (2020).

Principle 1: *“No action taken by law enforcement agencies, persons employed within those agencies, or their agents should change data which may subsequently be relied upon in court.” Horsman, G. (2020).*

This principle states that no forensics investigator or any member of law enforcement agency should change the original data as this evidence may be needed in the court – this will help maintain integrity and chain of custody maintaining clear records to show its trustworthy when presenting in court.

I have applied this principle in my forensics investigation by making a disk image of Tracy’s hard drive this will help keep original hard drive safe and prevent any accident changes being made.

Principle 2: *“In circumstances where a person finds it necessary to access original data, that person must be competent to do so and be able to give evidence explaining the relevance and the implications of their actions.” Horsman, G. (2020).*

This principle states that in an event of requiring to access the original digital evidence e.g., turning on suspects laptop or opening/browsing through certain file on the hard drive the person investigating this must know what they are doing and be appropriately be trained in digital forensics investigating they must clearly be able to explain why they had to access the original data and what impact did it had, this is to prevent altering or damaging original data or digital evidence as this could cause the case being thrown out from the court due to weaker evidence

I am adhering to this principle by making sure that I am professional in terms of how I access the original data making sure that I do not alter any data alternatively I can also document my process and what I did throughout accessing the original data.

Principle 3: “An audit trail or other record of all processes applied to digital evidence should be created and preserved. An independent third party should be able to examine those processes and achieve the same result.” Horsman, G. (2020).

This principle states that every step analysing the digital evidence should be recorded and stored appropriately while an independent organisation should be able to look at your records and findings and able to follow exact steps this adds transparency and accountability when handling digital evidence.

When handling Tracy’s external hard drive I will make sure to keep record of what tools and software I used in what OS as well as keeping track of date and time

Principle 4: “The person in charge of the investigation has overall responsibility for ensuring that the law and these principles are adhered to.” Horsman, G. (2020).

This principle means that the lead investigator has the responsibility to make sure relevant laws are followed such as Computer Misuse Act 1990, GDPR, Data Protection Act and CPIA Act 1996 while also making sure that all ACPO guidelines are meant so that it keeps consistency and accountability across the investigation team/department.

1.5 Software and tools

Type	Software
Operating System Details	Edition Windows 11 Home Version 24H2 Installed on 08/01/2025. OS build 26100.3775 Experience Windows Feature Experience Pack 1000.26100.66.0
Forensics platform	AUTOPSY Digital Forensics Version 4.22.0 for Windows https://www.autopsy.com/ 

INVESTIGATION

2.1 Items Analysed

A compressed file that contains bit-by-bit information of the entire content in Tracy's external hard drive was created known as Disk Image this will prevent any accidentally changes being made to the original data.

The following Disk Image will be analysed for forensics investigations:

- [tracy-external-2012-07-16-final. E01](#)
- VM. vmdk_276

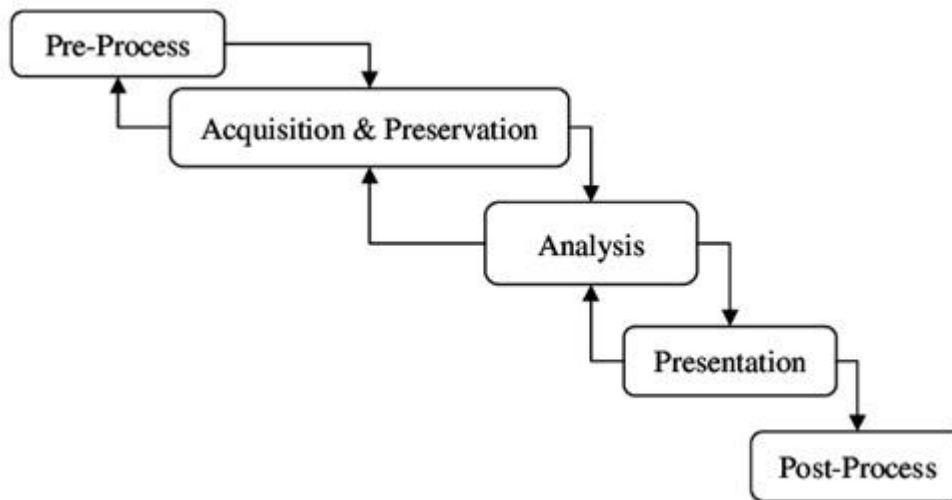
External Hard Drive File Metadata

Name	/img_tracy-external-2012-07-16-final.E01
Type	E01
MD5	1c68890eb70449fbb7fcf205740faa
Sector Size	512
Description	Flash Drive

VM.vmdk_276

Name	CORAL-PC
Program Name	Windows 7 Professional Service Pack 1
Path	C:\Windows
Description	OS Drive (Windows 7 Professional Service Pack 1)

2.2 Integrity Preservation



[\(Forensics framework\)](#)

Preserving digital evidence is an important part in digital forensics this will maintain the integrity of digital evidence when presenting in the court, to preserve digital evidence the authorities will secure the scene by isolating devices and make sure are devices e.g., Laptop, PC, mobile phone, USBs, hard drives, Digital and security cameras this will prevent the suspect from destroying or erasing the data, each digital device must be labelled and chain of custody should be followed

To maintain integrity of disk image file I will need to generate a hash as this verifies integrity of the data, any changes being made to disk image will result in a different hash value
Suhartono, J. (2023)

Tracy External

Name	Type	Size (Bytes)	Sector Size (Bytes)
 tracy-external-2012-07-16-final.E01	Image	14134280192	512

Hex	Text	Application	File Metadata	OS Account	Data Artifacts	Analysis Results	Cont
Metadata							
Name:		/img_tracy-external-2012-07-16-final.E01					
Type:		E01					
Size:		14134280192					
MD5:		1c68890eb70449fbb7fcf205740faa					
SHA1:		Not calculated					
SHA-256:		Not calculated					
Sector Size:		512					
Time Zone:		Europe/London					
Acquisition Details:		Description: Tracy's External					
:		Notes: Imaged on 2012-07-16 at 11:17:45 EST					
:		Acquired Date: Mon Jul 16 16:17:45 2012					
:		System Date: Mon Jul 16 16:17:45 2012					
:		Acquiry Operating System: Linux					
:		Acquiry Software Version: 20100226					
Device ID:		1700a86b-30e9-41fe-afe3-e6f8af1b9551					
Internal ID:		1					

(Figure 1)

2.3 Methodology

I will be using Autopsy Digital forensics software to scan and gather evidence, the evidence I will be looking for.

- Email/messages between Tracy and Carry as well as Pat.
- Any relevant information about stamps
- Any relevant information showing Tracy was about to commit theft.

Autopsy is reliable free to use software using by professionals, figure 2 shows the modules used while importing [tracy-external-2012-07-16-final.E01](#)

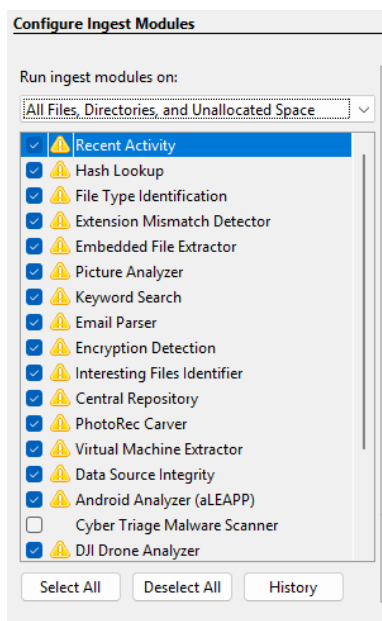
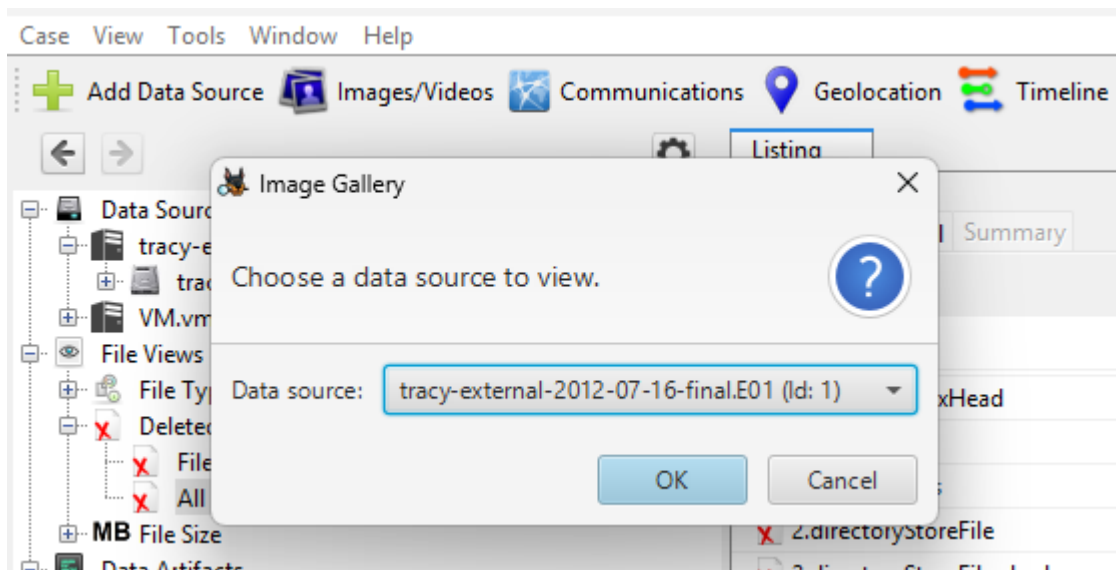


Figure 2

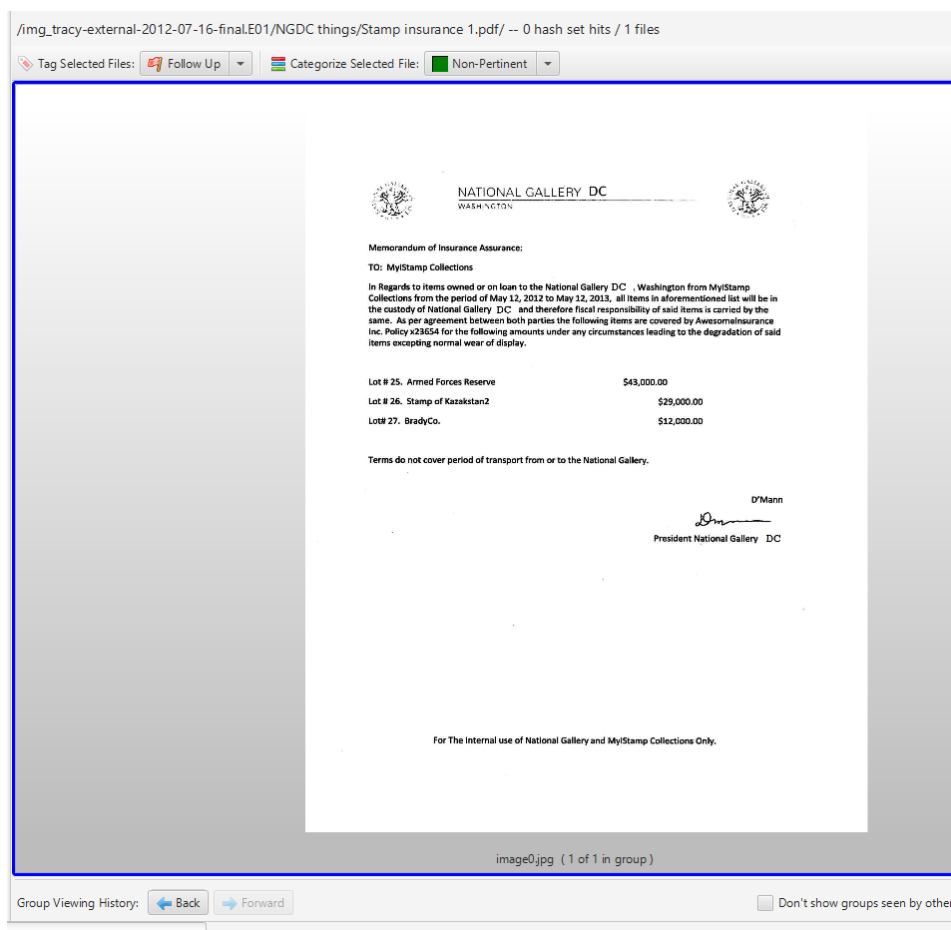
Figure 2 shows the plugins that analyse data from a data source such disk images, they perform various useful task like keyword searching, recovering data and file filtering.

2.4 Investigation Results and Analysis

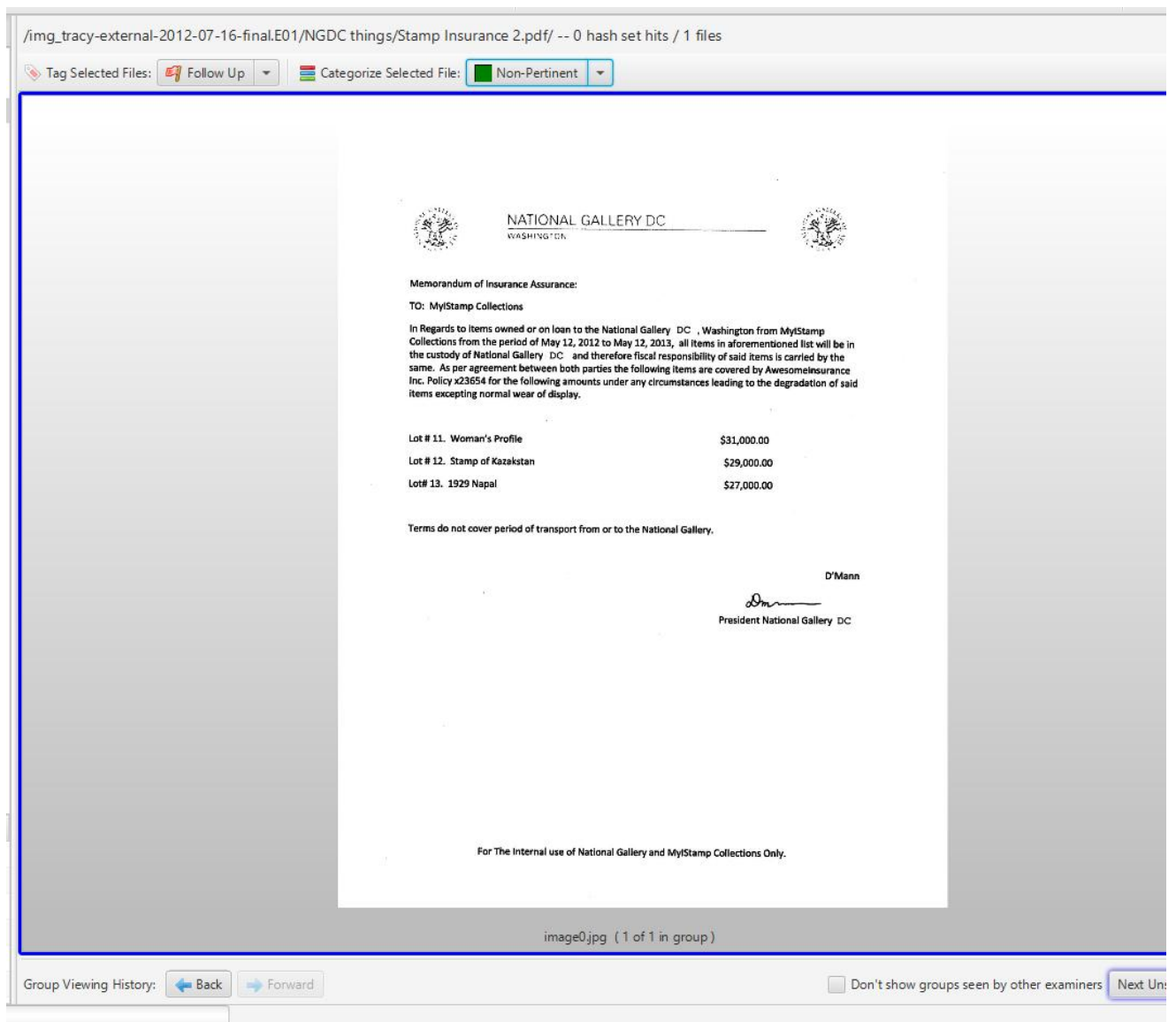
2.4.1 Stamp Insurance



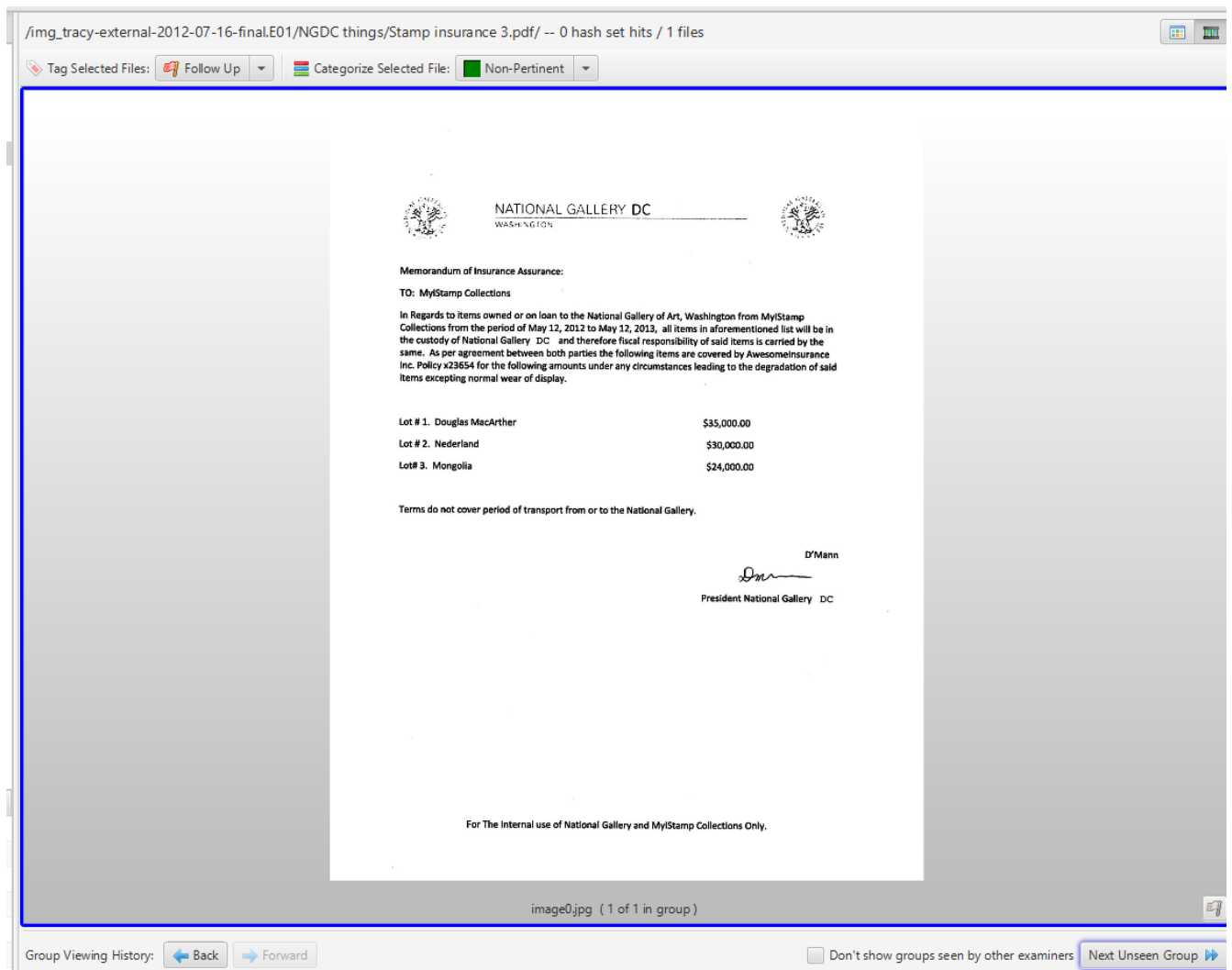
(Figure 3)



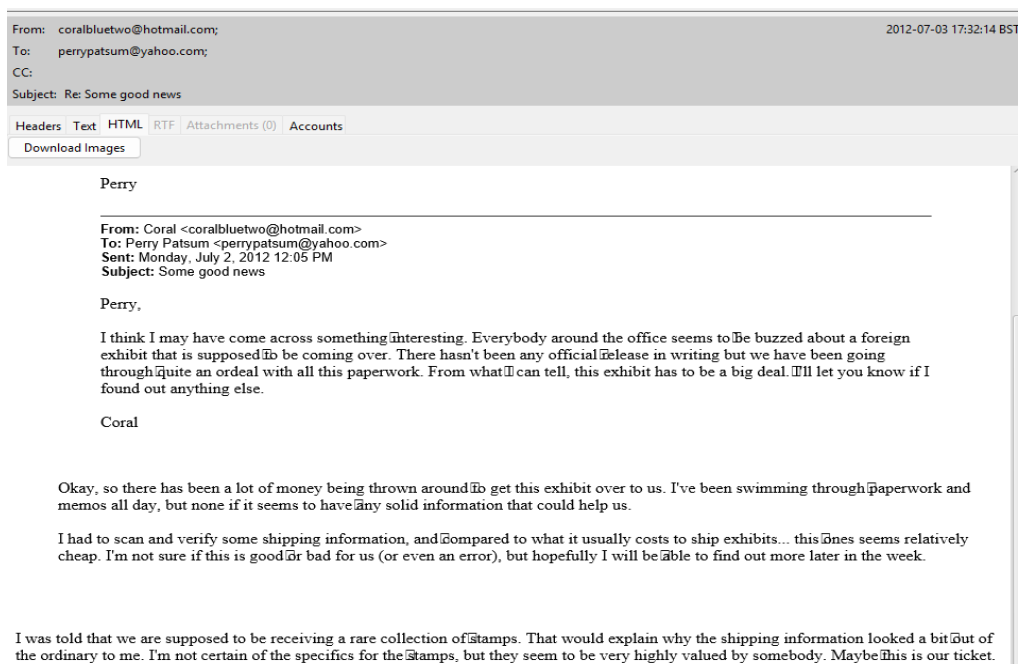
(Figure 4)



(Figure 5)



(Figure 6)



(Figure 7)

In Figure 3 I used *Image/videos* feature in Autopsy to see all the image and video file formats this include PNG, JPG, Jpeg, MP4, etc, as you can see Figure 4 – 6 shows stamp insurance and delivery information this suggests that Tracy may have been interested in stamps as she was looking through the paper work and actively searching for more information about the stamps. “*They seem to be very highly valued by somebody. Maybe this is our ticket.*” Said by Tracy shows clear intent to steal the stamps as they are rare and extremely valuable.

2.4.2 Web search

Below image shows Tracy has searched about stamps on multiple occasions this proves further that she was interested in the stamps and therefore intended to steal them as they are valuable

Web Search								
Term:	value+of+international+stamps							
Time:	2012-07-05 21:24:23 BST							
Domain:	dmtracker.com							
Program Name:	Internet Explorer Analyzer							

Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
index.dat				bing.com	thunderbird	Internet Explorer Analyzer	2012-06-28 22:48:10 BST	VM.vmdk
index.dat				bing.com	value of international stamps	Internet Explorer Analyzer	2012-07-05 21:23:56 BST	VM.vmdk
index.dat				bing.com	rare stamps	Internet Explorer Analyzer	2012-07-05 21:24:20 BST	VM.vmdk
index.dat				bing.com	thunderbird	Internet Explorer Analyzer	2012-06-28 22:48:11 BST	VM.vmdk
index.dat				bing.com	thunderbird	Internet Explorer Analyzer	2012-07-05 21:23:51 BST	VM.vmdk
index.dat				bing.com	rare stamps	Internet Explorer Analyzer	2012-07-09 17:17:12 BST	VM.vmdk
index.dat				bing.com	thunderbird	Internet Explorer Analyzer	2012-06-28 22:48:10 BST	VM.vmdk
index.dat				bing.com	rare stamps	Internet Explorer Analyzer	2012-07-05 21:24:20 BST	VM.vmdk
index.dat				bing.com	rare stamps	Internet Explorer Analyzer	2012-07-05 21:24:20 BST	VM.vmdk
index.dat				bing.com	thunderbird	Internet Explorer Analyzer	2012-06-28 22:48:11 BST	VM.vmdk
index.dat				bing.com	thunderbird	Internet Explorer Analyzer	2012-07-05 21:23:51 BST	VM.vmdk
index.dat				bing.com	rare stamps	Internet Explorer Analyzer	2012-07-12 18:57:37 BST	VM.vmdk
index.dat				dmtracker.com	value+of+international+stamps	Internet Explorer Analyzer	2012-07-05 21:24:23 BST	VM.vmdk

2.4.3 Coral Alias

Through looking at web history data, we can see below cookies detail in figure 9 by clicking on file [coral@lijit\[1\].txt](#) we can see the address links back to [tracysumtwelve@gmail.com](#) and that Coral is temporary name used by Tracy to talk to Pat who we can assume uses alias Perry as both of them communicated on messages/email. Using aliases is clear indication of doing something illegal as they are avoiding to be not identified.

Basic Properties

Login: coral

Full Name:

Address: S-1-5-21-1664576105-2049440411-488585963-1000

Type:

Creation Date: 2012-06-27 23:53:00 BST

Object ID: 475

(Figure 8)

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
coral@kontera[1].txt			0	2012-07-05 22:24:29 BST	2012-07-05 22:24:29 BST	2012-07-05 22:24:29 BST	2012-07-05 22:24:29 BST
coral@lijit[1].txt			0	2012-07-05 22:24:36 BST	2012-07-05 22:24:36 BST	2012-07-05 22:24:36 BST	2012-07-05 22:24:36 BST

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 1 of 1 Result < >

Cookie Details

Domain: google.com

URL: mail.google.com/mail/u/0

Name: gmailchat

Value: tracysumtwelve@gmail.com/194438

Program Name: Internet Explorer Analyzer

Dates

Created: 2012-07-12 17:04:19 BST

Source

Host: VM.vmdk_276 Host

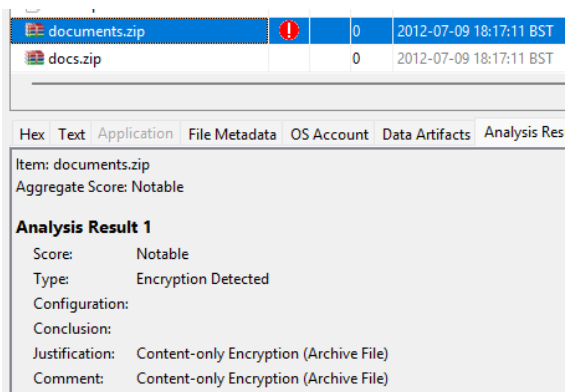
Data Source: VM.vmdk

File: /img_VM.vmdk/vol_vol3/Users/Coral/AppData/Roaming/Microsoft/Windows/Cookies/Low/coral@mail.google[1].txt

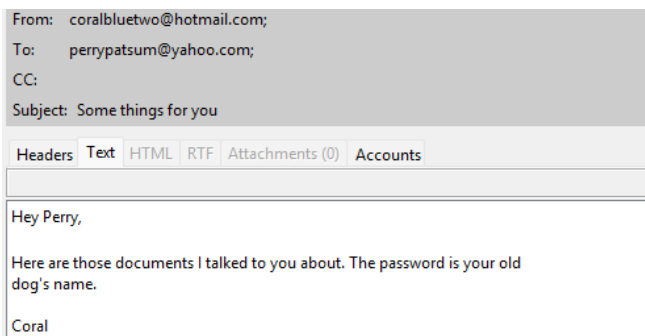
(figure 9)

2.4.3 Encrypted documents

From figure 10 we can see Tracy has encrypted documents.zip which contains 3 stamp insurances we can also see that in figure 11 Coral known as Tracy sent these encrypted documents to Perry who is her brother Pat this clearly shows that Tracy had intent to steal the stamps as the documents where quite important and she intentionally encrypted the documents so that only the right person could access it, in this context Pat is the right person and he likely knew the password.



(Figure 10)



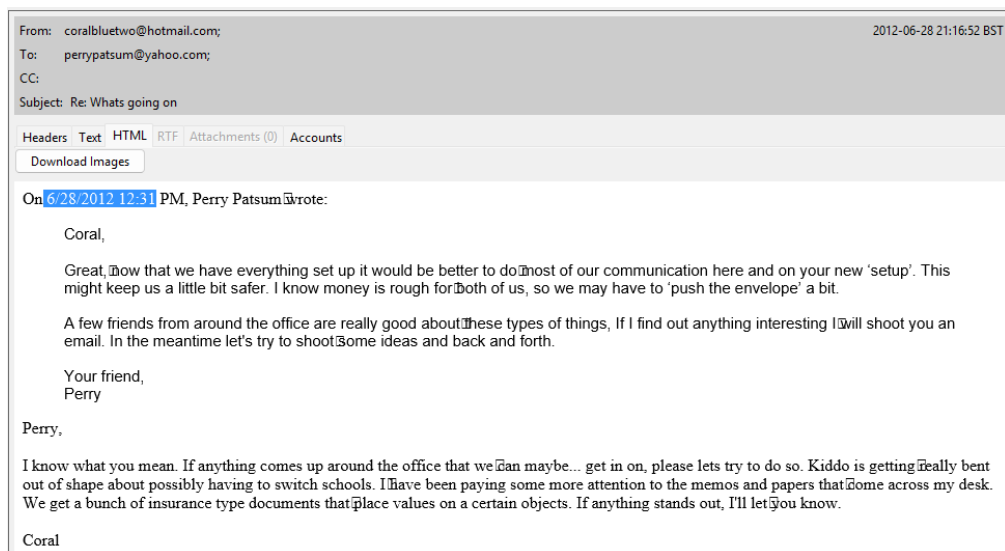
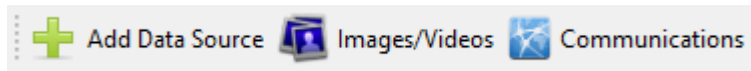
(Figure 11)

Listing				
/img_VM.vmdk/vol_vol3/Users/Coral/Documents/documents.zip/docs				
Table Thumbnail Summary				
Name	S	C	O	Modified Time
 .DS_Store			1	2012-07-09 14:42:58 BST
 Stamp insurance 1.pdf			2	2012-07-06 14:39:52 BST
 Stamp Insurance 2.pdf			2	2012-07-06 14:39:52 BST
 Stamp insurance 3.pdf			2	2012-07-06 14:39:52 BST

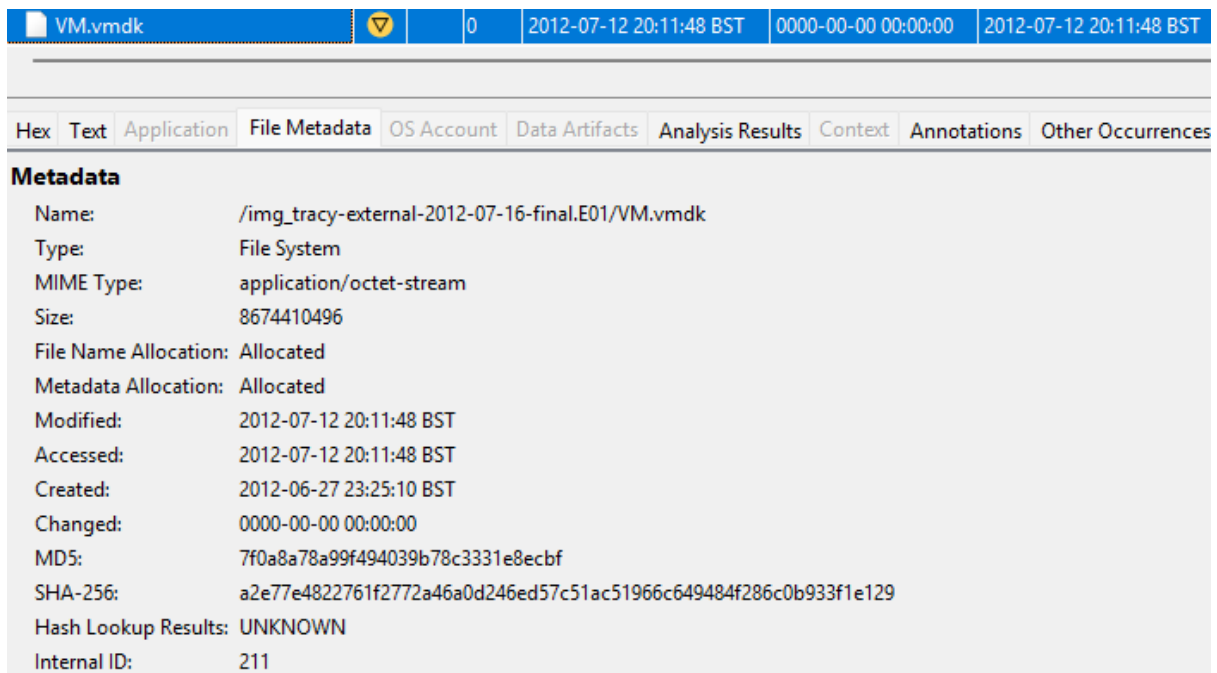
(Figure 12)

2.4.4 Use of Virtual Machine

Through using the Communications tool, I have discovered this email (figure 13) which talks about new setup to keep them safe, this email was sent on 6/28/2012 – (UK format 28/6/2012) at 12:31 PM where as the new setup potentially Virtual Machine in figure 14 was created 13 hours prior to this email this suggest that Tracy might wanted to keep this messages and communication hidden as VMs can easily be discarded/deleted making it harder to recover evidence. The act of keeping it hidden shows they know they are involved in planning to steal stamps and making sure to not get caught.



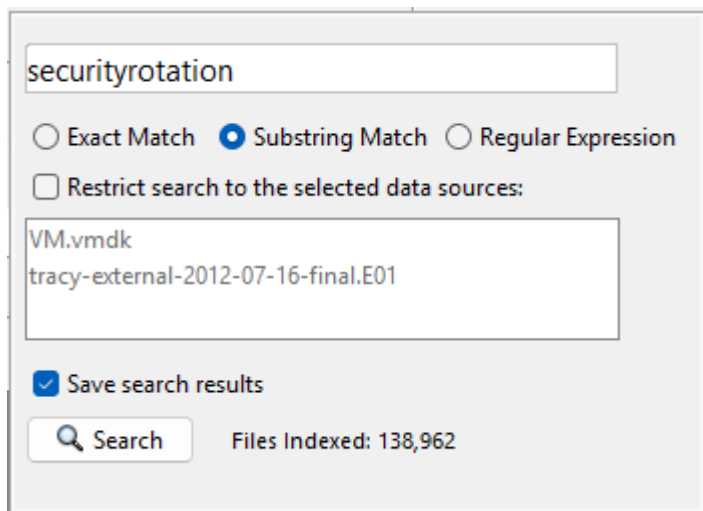
(Figure 13)



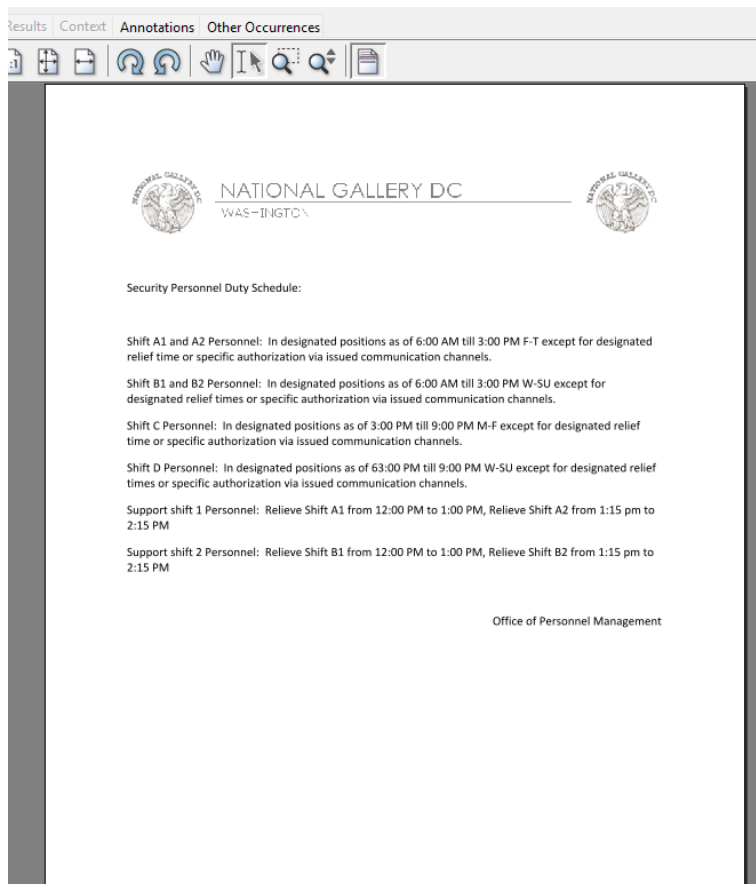
(Figure 14)

2.4.5 Breach of Confidentiality

Through using the Search Keyword feature on Autopsy I have discovered that Tracy had sent Security rotation document to Carry who is Karsonovian and was looking to organise flash mob for National Art Gallery. The security rotation document in figure 16 is highly sensitive not only it shows what time each shift is but also shows number of security personnel this means that this is important document that must not be shared with anyone outside of the organisation, Tracy has clearly leaked this document to Carry as seen in figure 17 tracysumtweleve@gmail.com (Tracy's email address) she attached securityrotation.pdf while also insisting that no one else sees this clearly shows that she knows she is doing something illegal which suggests she would also attempt steal the rare stamps

The image shows a screenshot of the Autopsy search interface. At the top, there is a search bar containing the text "securityrotation". Below the search bar, there are three radio buttons for search criteria: "Exact Match", "Substring Match" (which is selected), and "Regular Expression". Below these, there is a checkbox labeled "Restrict search to the selected data sources:" which is currently unchecked. Under this checkbox, there is a list of data sources: "VM.vmdk" and "tracy-external-2012-07-16-final.E01". At the bottom left, there is a checkbox labeled "Save search results" which is checked. To the right of this checkbox is a "Search" button with a magnifying glass icon. To the right of the "Search" button, it says "Files Indexed: 138,962".

(Figure 15)



(Figure 16)

["
:"\u003cspan class\u003d\"yP\" email\u003d\"carysum2012@yahoo.com\" \"\u003eCary\u003c/span\u003e \"\u003cspan class\u003d\"yP\" email\u003d\"tracysumtwelve@gmail.com\" \"\u003eme\u003c/span\u003e (9)\" \"\u0026raquo;\u0026nbsp;\" \"Long time no see.\" \"Okay Carrie
!\u0026#39;m going to send this but you need to make sure no one else sees it okay I could get\",0,\" \"securityrotation.pdf\", \"Jul 11\", \"Wed, Jul 11, 2012 at 2:53 PM\", 1342033642316000, 1, [,
, 0,]

(Figure 17)

CONCLUSION

After investigating the hard drive, I believe Tracy had the intent to steal the stamps as she was quite interested in rare stamps, she also sent stamps insurance to determine the value of the stamps to her brother, she was also searching for more information about the stamps as seen in emails/messages by asking around in the office not only this she also sent confidential information about security to Carry which means she was involved in working with Carry who was going to organise the flash mob. Tracy also attempted to keep information hidden through:

- Using Virtual Machine when communicating with Pat to keep themselves safer shows that they had intentionally installed VM to keep information private and could be discarded afterwards to make it harder to recover.
- Use of Aliases by Tracy and Pat suggests that they did not want to be identified as they are stealing highly valuable stamps – this also shows they had planned of committing the crime.
- Tracy Encrypted the documents is highly suspicious as she had stamps insurance information on them which was sent to Pat making it clear they was going to steal the stamps as the insurance suggested the stamps are highly valuable.

REFERENCES

No.	Reference
1	<i>ACPO Guidelines & Principles explained</i> (no date) <i>Forensic Control</i> . Available at: https://forensiccontrol.com/guides/acpo-guidelines-principles-explained/ (Accessed: 15 April 2025).
2	Horsman, G. (2020) ACPO principles for Digital evidence: Time for an update?, <i>Forensic Science International: Reports</i> . Available at: https://www.sciencedirect.com/science/article/pii/S2665910720300220 (Accessed: 15 April 2025).
3	(No date) <i>The feasibility of creating a Universal Digital Forensics Framework</i> . Available at: https://www.heraldopenaccess.us/openaccess/the-feasibility-of-creating-a-universal-digital-forensics-framework (Accessed: 17 April 2025).
4	<i>Suhartono, J. (2023) Process in collection and preservation in digital forensic investigation, School of Information Systems</i> . Available at: https://sis.binus.ac.id/2023/11/03/process-in-collection-and-preservation-in-digital-forensic-investigation/#:~:text=Calculate%20cryptographic%20hash%20values%20(e.g.%2C%20MD5%2C%20SHA%2D1%2C,integrity%20of%20the%20evidence%20throughout%20the%20inv estigation. (Accessed: 17 April 2025).

APPENDIX A

Autopsy Forensic Report

HTML Report Generated on 2025/04/19 22:34:42

Case: Tracy
External
Hard Drive

Case Number: 023

Number of
data sources in
case: 2

Examiner: Usman

Image Information:

tracy-external-2012-07-16-final.E01

Timezone: Europe/London

Path: C:\Users\usman\Downloads\tracy-external-2012-07-16-final.E01

VM.vmdk

Timezone: Europe/London

Path: C:\Users\usman\Desktop\Digital Forensics\Tracy External Hard
Drive\ModuleOutput\Virtual Machine Extractor\tracy-external-2012-
07-16-final.E01_1_2025_04_08_00_03_41\1\VM.vmdk

Software Information:

Autopsy Version:	4.22.0
Android Analyzer Module:	4.22.0
Android Analyzer (aLEAPP) Module:	4.22.0
Central Repository Module:	4.22.0
DJI Drone Analyzer Module:	4.22.0
Data Source Integrity Module:	4.22.0
Email Parser Module:	4.22.0
Embedded File Extractor Module:	4.22.0
Encryption Detection Module:	4.22.0
Extension Mismatch Detector Module:	4.22.0
File Type Identification Module:	4.22.0
GPX Parser Module:	1.2
Hash Lookup Module:	4.22.0
Interesting Files Identifier Module:	4.22.0
Keyword Search Module:	4.22.0
PhotoRec Carver Module:	7.0
Picture Analyzer Module:	4.22.0
Recent Activity Module:	4.22.0
Virtual Machine Extractor Module:	4.22.0
YARA Analyzer Module:	4.22.0
iOS Analyzer (iLEAPP) Module:	4.22.0

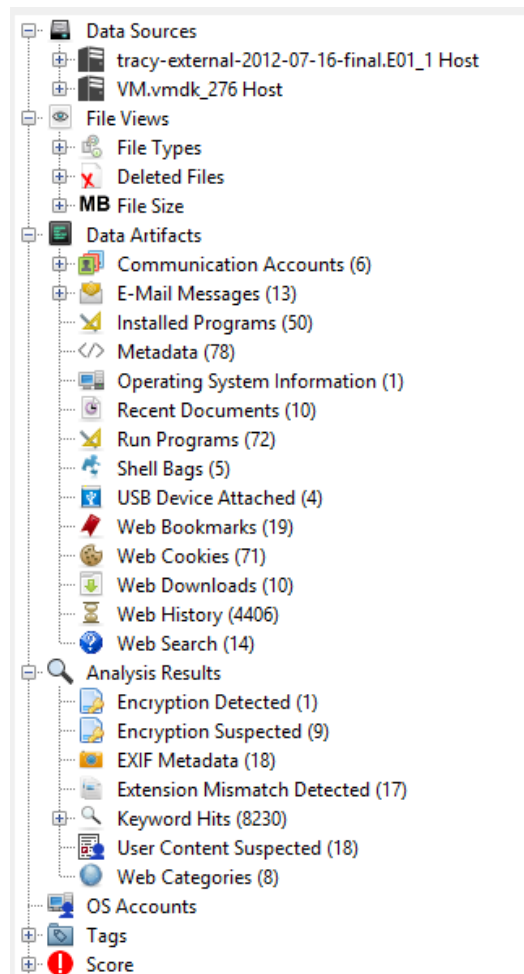
Ingest History:

Job 1:

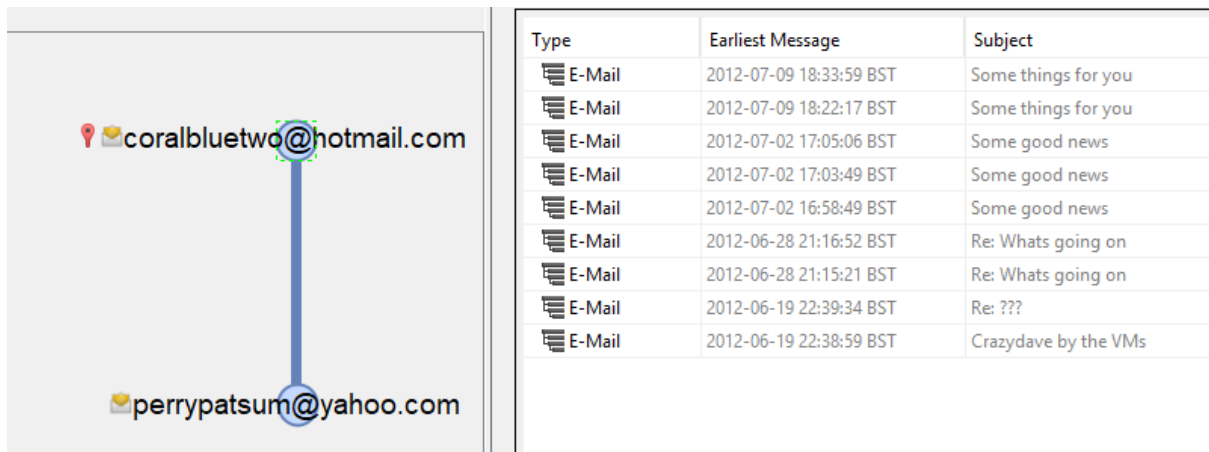
Data Source: tracy-external-2012-07-16-final.E01
Status: COMPLETED
Enabled Modules: Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Data Source Integrity
Android Analyzer (aLEAPP)
DJI Drone Analyzer
YARA Analyzer
iOS Analyzer (iLEAPP)
GPX Parser
Android Analyzer

Job 2:

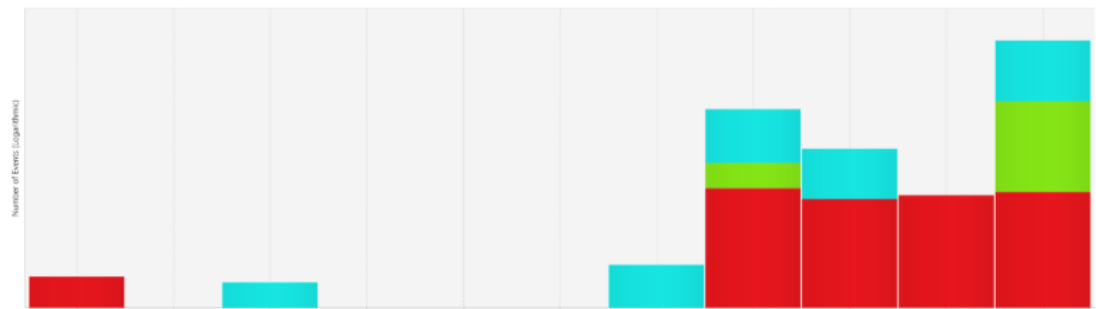
Data Source: VM.vmdk
Status: CANCELLED
Enabled Modules: Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
Data Source Integrity
Android Analyzer (aLEAPP)
DJI Drone Analyzer
YARA Analyzer
iOS Analyzer (iLEAPP)
GPX Parser
Android Analyzer



APPENDIX B



Timeline Snapshot



Time Range: Friday, 18 October 2002 at 08:39:06 British Summer Time to Friday, 13 July 2012 at 10:40:45 British Summer Time

☒ Limit event types to

☒ File System

☒ File Accessed

☒ File Changed

☒ File Created

☒ File Modified

